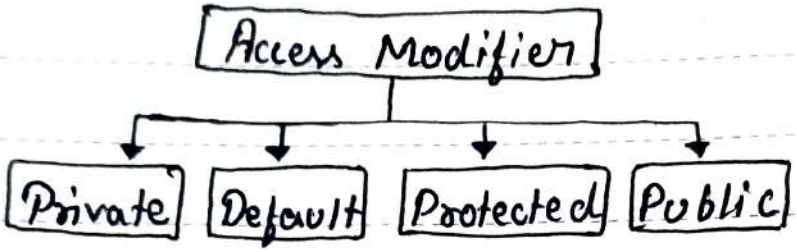
Access Modifiers in Java :-

*) Access Modifiers defines how variables, methods, classes can be accessed from other parts of our program.



Private Access Modifier :-

-> accessible only within the class in which they are declared.

```
class Person {
        private String name;
        public void setName (String name) {
            this.name = name; // accessible within class
        }
        public String getName () {
            return name;
        }
    }
```

```
public class Geeks {
    public static void main (String[] args)
    { Person p = new Person();
      p.setName ("Allice");
      sout( p.name); // Error :- 'name' has
                                 private access

      sout (p.getName()); // but can be accessed
                             by methods, getters
                             or setter if method
                             is not private.
    }
}
```

-> Direct access to 'name' is not allowed outside 'Person' class, enforcing encapsulation.

-> But can be accessed using non-private getters or setters.

Default Access Modifier :-

-> If no access specifier is specified, the member is default (package-private) access and can only be accessed within the same package.

-> Which means only classes within same package can access it.

file : 1

```
Package p1;
class Greek {  // class Greek is default
    void display ()
    {  sout("Hello world");
    }
}
```

Greeks.java

file = 2

```
Package p2;
Import p1.*;  // importing package p1, without importing
                we can't access other package members.
class GreekNew {  // default GreekNew class
    public static void main (String args[])
    {  Greek o = new Greek();  // Accessing from p1 using
                                  object of class Greek.
       o.display ();
    }
}
```

GreekNew.java

Protected Access Modifier :-

*) Can be accessable only within the same package.

*) Cannot access in another package even after importing also.

*) But can be accessed in another package if

that another package contains sub-class of the class present in main package.

file: vehicle.java

```
package P1;
public class Vehicle {
    protected int speed;
}
```

file : Bike.java

```
package P2;
import p1.Vehicle;
public class Bike extends Vehicle
{ void showspeed() {
        speed = 100; //allowed : subclass in different package
        Sout (speed);
}  }
//Public class car
{
    speed = 100; // error, cannot access in non-sub class;
}
```

This is neccessary

file : Test.java

```
Package P2;
Import P1.vehicle;
Public class Test {
    PSVM() { Bike b = new Bike();
             b.showSpeed(); //100
```

~~Vehicle v = new~~

```
Vehicle v = new Vehicle();
  Sout (v.speed); //Error : cant access protected
                                outside package & non-subclass
```

Public Access Modifier :-

*) Accessible from everywhere in Program.

*) No restriction on scope of public data members

```
class MathUtils {
    Public static int add (int a, int b) {
        return a+b;
    }
}

Public class Main {
    Public static void main (String[] args) {
        Sout (MathUtils.add (5, 10)); //accessable
                                        anywhere

        Sout (add (5, 10)); //error, should used
                                class name.
    }
}
```

Top-level classes or interfaces can not be declared as private because private means "Only visible with the enclosing class".